

Estimating the size of a population

Answers

Population of nettles in a paddock

1. A paddock has been invaded by stinging nettles. The quadrat method was used to estimate the population of nettles. A 100 m by 100 m area was divided into 100 smaller quadrats. Each quadrat measured 10 m by 10 m. The number of nettles in 15 of these random quadrats was measured. The results are shown in the table below.

Quadrat number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Number of nettles	4	6	3	2	0	4	2	3	1	4	4	2	3	1	6

- (a) Calculate the average number of nettles per quadrat.

$$\frac{45}{15} = 3.0$$

- (b) Calculate the total area investigated.

$$100 \times 100 = 10\,000 \text{ m}^2$$

- (c) Calculate the abundance of nettles in the whole paddock by using the formula:

$$\text{Total number} = \frac{\text{average number per quadrat} \times \text{total area}}{\text{area of quadrat}}$$

$$\frac{3.0 \times 10\,000}{10 \times 10} = 300$$

- (d) Estimate the density of nettles in the paddock by dividing your answer to part (c) by the total area in square metres.

$$\frac{300}{10\,000} = 0.030 \text{ nettles/square metre}$$

Capture–recapture

2. A biologist captures 14 goannas in a 10 square kilometre area of open woodland. He tags them and releases them back into the woodland. Six months later he returns and captures 10 goannas of which 7 have tags. Estimate the total population of goannas in that area using the following formula:

$$\text{Total number} = \frac{\text{number of animals on first visit} \times 100}{\text{average percentage of tagged animals per recapture}}$$

$$\text{Average percentage of goannas per recapture} = \frac{7}{10} \times 100 = 70\%$$

$$\text{Total population of goannas} = 14 \times \frac{100}{70} = 20$$